

A microwell platform for high-throughput longitudinal phenotyping and selective retrieval of organoids

Highlights

- Easy-to-manufacture microwell device for single-organoid phenotypic profiling
- Proof-of-principal characterizations of various phenotypic traits
- Organoid retrieval for genotype-phenotype matching
- Companion imaging analysis suite for seamless integrative analysis

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In brief

The manuscript describes a microwell platform for phenotypic profiling of organoids at single-cell/organoid resolution. Various phenotypic traits such as growth rate, movement, and apical-basal polarity have been profiled. Coupled with the mechanism for physical organoid retrieval and single-organoid sequencing, the microwell platform enables genotype-phenotype mapping of phenotypes-of-interest.



Article

A microwell platform for high-throughput longitudinal phenotyping and selective retrieval of organoids

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SUMMARY

Organoids are powerful experimental models for studying the ontogeny and progression of various diseases including cancer. Organoids are conventionally cultured in bulk using an extracellular matrix mimic. However, bulk-cultured organoids physically overlap, making it impossible to track the growth of individual organoids over time in high throughput. Moreover, local spatial variations in bulk matrix properties make it difficult to assess whether observed phenotypic heterogeneity between organoids results from intrinsic cell differences or differences in the microenvironment. Here, we developed a microwell-based method that enables high-throughput quantification of image-based parameters for organoids grown from single cells, which can further be retrieved from their microwells for molecular profiling. Coupled with a deep learning image-processing pipeline, we characterized phenotypic traits including growth rates, cellular movement, and apical-basal polarity in two CRISPR-engineered human gastric organoid models, identifying genomic changes associated with increased growth rate and changes in accessibility and expression correlated with apical-basal polarity.

A record of this paper's transparent peer review process is included in the supplemental information.

INTRODUCTION

The advent of three-dimensional (3D) organoid culture methods that recapitulate tissue structure, multi-lineage differentiation, pathology, and disease phenotypes while retaining the tractability of *in vitro* systems has revolutionized the study of various diseases, including cancer.^{1–6} Key strengths of organoids over conventional transformed two-dimensional (2D) cell lines for studying cancer include the greater similarity of 3D versus 2D models to *in vivo* settings and the relatively clean genomic background of healthy primary organoids, which can be engineered via CRISPR-Cas9 editing or other techniques to harbor alterations in oncogenes or tumor suppressors that promote malignant progression.^{7–9} Several studies have demonstrated

the transformation of normal colon, stomach, and pancreas organoids into invasive carcinomas via the simultaneous introduction of multiple oncogenic “hits.”^{7,8,10–12} These minimally transformed forward-genetic models have yielded insights into the requirements for transformation and their tissue-specific functional consequences from multi-omics genetic data. However, high-throughput phenotypic characterizations of these 3D organoid models are lacking.¹³ To date, the phenotypic characterization of single organoids has remained challenging, as most techniques suitable for segmenting cells and quantifying the growth of 2D cell cultures do not readily extend to 3D cultures, and many methods were also designed for bulk populations.^{14–17} In addition, the most common strategy for culturing organoids in 3D involves resuspending dissociated cells in

Matrigel or Cultrex basement membrane extract (BME), which are commercially available extracellular matrix (ECM) mimics.¹⁸ In this standard method, organoids are embedded throughout the full height of the ECM such that they are not in focus within a single imaging plane. As a result, imaging organoids requires multiple images at different heights, and image quality is degraded by scattering within the sample (unlike with 2D cell cultures where a single z axis focal plane [or maximally several] is required to capture the morphological features of different cells¹⁹) (Figure S1). In addition, organoids within a bulk ECM can be formed from clusters of aggregated cells rather than from a single cell that expands independently, making it difficult to determine whether an observed phenotypic trait reflects variable starting cell population size or is intrinsic to individual cells at the start of organoid growth.^{20,21} Distinguishing intrinsic from extrinsic heterogeneity is further complicated by variations in the organoid's growth environment, as organoids seeded close to other organoids may be affected by cell-cell paracrine or juxtacrine signaling^{22,23} and organoid position relative to the margins of the ECM hemisphere can impact the diffusion of growth factors and/or drugs.^{24,25} Finally, the same observed bulk growth differences could result from differences in median organoid growth rate²⁶ or the fraction of cells that grow, hindering the interpretation of such data.

Microfabricated microwells represent an alternative to bulk culture methods and have been used extensively in the materials science and tissue engineering fields for 3D cell culture.^{15,20,27–30} Various microwell designs in the literature have been introduced to overcome specific challenges pertaining to the image-based analysis of organoids in bulk cultures. Some devices facilitate high-content analysis of single or few organoids with the aim of understanding tissue regeneration and 3D cellular architecture but cannot profile large numbers of organoids.^{31–33} These designs also generally require expensive equipment (e.g., confocal and/or light sheet microscopy) and complex device fabrication procedures. By contrast, other microwell designs profile phenotypic features (e.g., whole-organoid morphology, growth rate, or movement) via simple bright-field or epifluorescence microscopy for many organoids at once. However, many of these methods rely on cellular aggregation to generate spheroids^{20,27,29,34–39} rather than growing organoids from a single cell, precluding investigations into underlying cellular and phenotypic heterogeneity. In addition, many existing microwell methods do not have an integrated analytical pipeline or require independent user implementation of analytical software into the overall workflow.^{29,40–42} Finally, most platforms lack the ability to selectively retrieve organoids of interest for downstream investigation, hindering linkage of phenotype and genotype.²⁸

Here, we present an open-source microwell platform and image-processing pipeline that enables the characterization of a variety of phenotypes for thousands of single organoids in parallel under near-identical conditions. These microwells are easy to fabricate and integrate into traditional cell culture workflows and can be imaged using standard inverted microscopes. We further demonstrate selective retrieval of specific organoids of interest for downstream characterization to link phenotype and genotype. Applying this new microwell platform to two established CRISPR-engineered organoid models of gastric tumorigenesis, representing pre-malignant (TP53 single gene knockout—TP53KO) and

malignant (TP53/ARID1A double knockouts [DKO]) states,^{7,43} we quantified cell size and position over time to determine single-organoid growth rates and migration patterns via time-lapse imaging and a neural network-based image analysis module^{44,45} for nearly 100,000 cell trajectories and over 8,000,000 microwell images. By fluorescently labeling nuclei and actin in both engineered organoid lines, we determined the proportion of cells with disorganized apical-basal polarity, a hallmark of malignant transformation,^{46,47} and demonstrated an enrichment in the DKO relative to the TP53KO model. We then retrieved individual DKO organoids with organized versus disordered polarity and performed dual chromatin accessibility and transcriptomic profiling, which implicated changes in accessibility among cell adhesion genes as altering apical-basal polarity. This microwell platform and associated analysis pipeline can readily be integrated within existing culturing protocols to investigate molecular and phenotypic features across organoids derived from diverse tissues (Figure S2).

RESULTS

Organoid models of gastric tumorigenesis and growth rate measurements in bulk culture

Here, we used two established organoid models of gastric tumorigenesis based on single and combinatorial gene editing via biallelic inactivation of the tumor suppressor *TP53* (TP53KO) or dual inactivation of *TP53* and *ARID1A* (DKO)⁷ (Figure 1A). The *TP53/ARID1A* DKO exhibited malignant phenotypes, including mucinous metaplasia and capacity for *in vivo* tumor growth upon xenotransplantation in mice, implying that this line has undergone malignant transformation.⁷ In contrast, the TP53KO gastric organoids exhibit normal morphology and are not tumorigenic even after long-term *in vitro* evolution despite harboring a constellation of copy-number variants associated with gastric cancer, implying that this line mimics a pre-malignant state.^{7,43} These defining features render these excellent models in which to perform systematic and quantitative characterization of additional phenotypes within and between models, including growth rates and the prevalence of apical-basal polarity.

As reported previously,⁷ images from bulk cultures grown in Cultrex BME revealed qualitative phenotypic differences between these gastric organoid lines, with increasing cellular disorganization for organoids lacking either *TP53* or both *TP53* and *ARID1A* (Figure 1A). Changes in cell growth, assessed by seeding bulk cultures with the same number of cells and then quantifying the fold change in the number of viable cells after 14 days of passaging in conditioned media, also suggested that DKO organoids were more proliferative than TP53KO organoids in bulk culture (Figure 1B), consistent with previous observations.⁷ However, these bulk culture methods could not determine the following: (1) if fold-change increases resulted from individual cells growing at a faster rate or from differences in the fraction of cells that grow, (2) whether the culture contained subpopulations with different growth behaviors, or (3) whether any observed subpopulations resulted from intrinsic differences between cells or cellular microenvironment (e.g., proximity to other organoids or to media surrounding the Cultrex BME boundary). Finally, organoids in bulk culture physically overlap and can merge with one another (Figure 1C), making it difficult

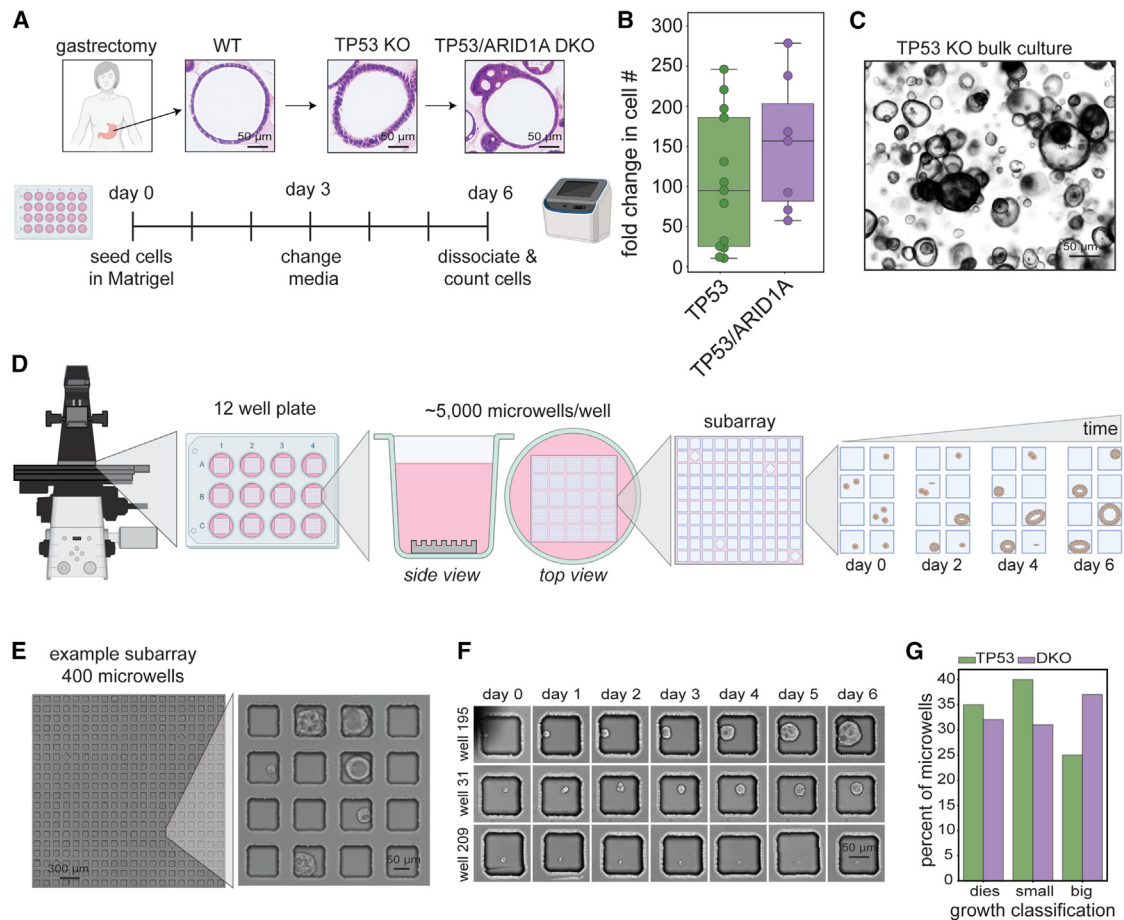


Figure 1. A microwell platform for high-throughput phenotyping of gastric cancer organoids

(A) Experimental pipeline and example histology images for sequential genome editing of TP53 and ARID1A in human gastric organoids, showing that loss of epithelial apical-basal polarity was associated with TP53/ARID1A DKO organoids. Organoid cultures were seeded in Cultrex BME and allowed to grow for 6 days before bulk measurements of growth and microscopy examination.

(B) Fold change in cell number from start to end of passage for TP53KO and DKO organoids ($n = 13$ and 7 , respectively); boxplots indicate median and upper/lower quartiles, whereas whiskers denote the full range of measurements; differences in cell number fold changes between cell types were not significant (as assessed by a Wilcoxon rank-sum test).

(C) Example bright-field image of TP53 organoids grown in bulk culture illustrating challenges with isolating individual organoids.

(D) Schematic of initial microwell experiments using bright-field microscopy. Each microwell array was placed onto a well of a 12-well tissue culture plate; single cells dissociated from organoids were seeded into the microwells and allowed to expand.

(E) Example image of 100 μ m microwell subarrays with organoids at the end of a 6-day time course.

(F) Example images showing growth trajectories of single cells seeded in microwells, demonstrating heterogeneity in growth patterns.

(G) Manual classification of growth trajectories for single cells by organoid size at end of time course (small, $<25\%$ of well occupied; big, $\geq 25\%$ of microwell occupied); a greater proportion of DKO organoids were classified as “big” compared with TP53KO (37% and 25%, respectively, from $n = 100$ each), although this is not statistically significant at a p value of 0.0824 (Fisher’s exact test).

to track the growth of individual organoids or isolate individual organoids of interest for downstream analysis.

Microwell arrays for high-throughput longitudinal phenotyping of single organoids

To address these challenges, we developed a microwell platform to perform time-resolved phenotyping of thousands of organoids in parallel under near-identical conditions (Figure 1D). Each single-layer polydimethylsiloxane (PDMS) microwell device contains arrays of 2,500–10,000 microwells (either $100 \times 100 \times 80$ or $200 \times 200 \times 80$ μ m, length $\times$ width $\times$ depth) placed directly at the bottom of each well (“macrowell”) of a 12-well culture

plate (Figures 1D and S3). Devices are fabricated by spin-coating PDMS onto master molds, thereby ensuring uniform thickness and enhancing image quality (Figure S4). To facilitate unique microwell indexing during downstream image processing, microwells are grouped within subarrays of 20×20 (100 μ m) or 10×10 (200 μ m) microwells, with a pattern of rotated microwells that uniquely identifies each subarray (Figure S5). The 100- μ m diameter microwells are optimized for high-throughput imaging of thousands of organoids in the same experiment (Figure S6), whereas 200- μ m diameter microwells are best suited for retrieving organoids with phenotypes of interest for single-organoid sequencing. All microwell devices were plasma treated and

coated with 0.5% BSA to render them hydrophilic (Figure S3), and our preliminary tests demonstrated that gastric organoids grown in these microwells recapitulated the composition and relative abundance of different cell types of the organoids grown in bulk ECM (Figure S7), although several notable differences in gene expression were observed in organoids cultured in either bulk manner versus those cultured in the microwell. Notably, the organoids cultured in the microwell have elevated expression in genes associated with cellular metabolic processes (Figure S7C).

Initial measurements of single-cell occupancy as a function of starting cell concentration for TP53KO cell lines established optimal loading concentrations of 6,000 cells/mL for 100- μ m microwells (26.15% of wells with a single cell) and 400 cells/mL for 200- μ m microwells (31.33% of wells with a single cell), respectively; the fraction of microwells containing a given number of cells was well fit by a Poisson distribution, consistent with expectations for stochastic loading (Figure S8).

For initial experiments, we seeded microwell arrays with single cells dissociated from TP53KO organoids and then imaged microwells daily via tiled bright-field imaging (Figures 1E and 1F). To visualize single organoid growth over time, we (1) stitched tiled images into a single image per macrowell per time point, (2) rotated stitched images to position microwell array edges parallel with image edges, (3) manually determined corner locations for each macrowell, and then (4) extracted individual microwell images from the rotated arrays by their position relative to the corner locations (Figure S5). We then manually inspected a subset of 100 microwells for each organoid line. Organoids grown in microwells appeared phenotypically similar to their bulk culture counterparts, with circular and cystic structures (Figures 1E and 1F). However, individual organoids grown from single cells under identical experimental conditions often showed dramatically different growth behavior: although some organoids showed rapid growth after seeding, defined as growth to $\geq 25\%$ of the microwell area (25/100 TP53KO; 37/100 DKO) (Figure 1F, top), others either grew very little ($< 25\%$ of the microwell area; 40/100 TP53KO; 31/100 DKO) (Figure 1F, middle) or showed signs of cell death/apoptosis (35/100 TP53; 32/100 DKO) (Figure 1F, bottom). An approximately equal proportion of TP53KO and DKO organoids exhibited growth; based on our manual classification, a greater proportion of DKO organoids showed larger and more rapid growth ($\geq 25\%$ of the microwell area) compared with TP53 (37% versus 25% of measured organoids, respectively) (Figure 1G).

High-throughput fluorescence imaging combined with deep-learning tracks thousands of cells in parallel

Bright-field microscopy made it possible to qualitatively assess organoid growth, but manual classification methods were both time consuming and confounded by a high degree of subjectivity. In addition, bright-field images lacked sufficient resolution to accurately determine whether observed growth was a result of cell division or simply an increase in organoid volume due to osmotic swelling of the lumen and the cells.^{48,49} To distinguish between these possibilities and gain information about nuclear localization within organoids, we engineered both TP53KO and DKO gastric organoid lines to express a nuclear fluorescent reporter by lentivirally inserting a mCherry-tagged copy of histone

2B (mCherry-H2B) (Figure 2A); in parallel, we profiled the distribution of cytoskeletal proteins using live-cell labeling of actin or tubulin (e.g., SiR Actin Kit from Cytoskeleton Inc.). To determine optimal conditions for fluorescence imaging and pilot new analysis pipelines, we loaded TP53KO- and DKO-engineered cells (at a concentration of 4,000 cells/mL) into 100- μ m microwells across three separate replicate experiments. The first and second experiments took place after 5 and 7 months of continuous passaging in conditioned media; the third experiment took place after ~ 8 months of continuous passaging, followed by a freeze-thaw cycle (required due to COVID-19 pandemic-related shutdowns) and an additional ~ 1 month of passaging post-freeze-thaw (Figure 2A). After loading, we mounted the entire plate assembly on an automated microscope with an incubation chamber and collected tiled images across the device in the bright-field, mCherry, and Cy5 (for fluorophore-tagged actin molecules) channels at 2-h intervals over 5 days for each experiment (Figure 2A). We then performed stitching, rotation, and microwell extraction for each imaging channel, similarly to the initial bright-field tests (STAR Methods). Time course image processing for each well yielded 8,014,800 total microwell images.

To efficiently extract information about the number, size, and relative positions of cells within this large imaging dataset, we turned to a neural network originally designed for optical microscopy-based image segmentation of cells grown in 2D monolayers.^{44,45} To test whether and under what conditions this deep-learning model can recognize and track cells within 3D organoids, we manually labeled individual mCherry-tagged nuclei positions in 2,711 images from the first experiment, used 2,169 of these images and a transfer learning approach to train an organoid-aware version of the model, and validated performance with the remaining 542 images (STAR Methods). To evaluate generalizability across experiments, we quantified model performance based on the number of accurately identified cells within an additional 912 manually labeled “test” images taken from all 3 experiments (306, 348, and 258 images, respectively) (Figure 2B). The model counts showed strong concordance with manual counts within the entire 5-day duration of each experiment ($R^2 = 0.83$ for the three combined experiments; Figure 2C; Figure S9 for per-experiment correlation). Precision and F1 scores were generally higher than recall for both validation and test sets, with the validation set displaying higher scores across performance metrics, as expected (Figures 2B and S9). Examination of error type revealed substantially more merges (in which 2 labeled cells were predicted by the model to be a single cell) compared with splits (where a single labeled cell was predicted to be 2 cells).

Further visual inspection of images showing regions identified as cells by the deep-learning model confirmed the accuracy of automated annotations over time (Figure 2D). Identifying all cells from microwells within a single macrowell required an average of 145 s for model processing per time point. Fitting the model-annotated cell area and cell number over time for each microwell to an exponential growth curve yielded quantitative growth rate estimates (Figure 2D); linearly interpolating the centroid position of single cells over time returned a lower bound for the distance traveled by a single cell prior to the first cell division (Figure 2E).

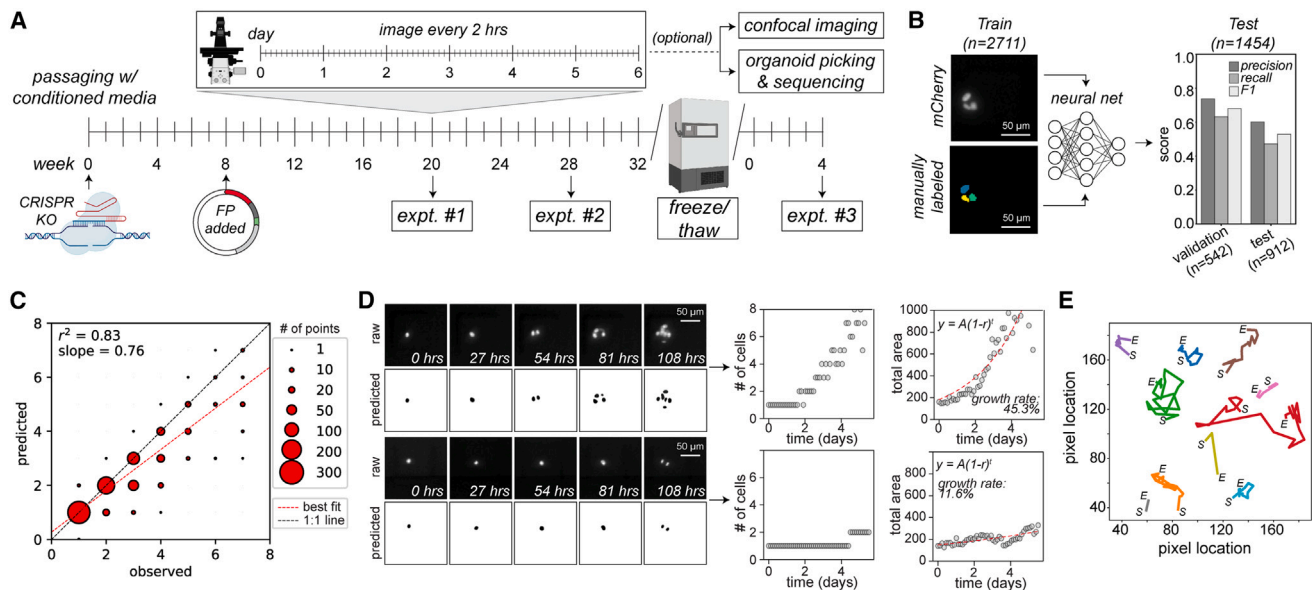


Figure 2. Neural net accurately identifies cells within fluorescence images to enable high-throughput image processing and organoid growth quantification

(A) Experimental timeline for cell engineering and imaging experiments.

(B) Pipeline for model training and validation (left) and performance metrics (right).

(C) Scatterplot comparing the number of model-predicted cells and manually annotated cells per microwell ($N = 912$ total microwells). Marker size indicates the number of events, dashed red line indicates linear regression, and dashed black line indicates 1:1 line.

(D) Example raw and model-predicted images (left) and model-annotated cell numbers and areas over time (right) for 2 microwells over 108 h of imaging. Red line shows a best fit exponential with the indicated equation and fit parameters; growth rate is reported based on the % daily change in area.

(E) Example single-cell position traces over time (0.72 pixels/ μm).

TP53/ARID1A DKO cells divide faster, but division rates decrease at higher occupancies

Using the trained deep-learning pipeline for 3D cultures, we then quantified organoid growth rates across all three initial fluorescence microscopy experiments (Figure 3A). In total, the 3 experiments profiled growth rates for 5,814, 1,238, and 3,679 loaded microwells, respectively, with the number of cells loaded per microwell following stochastic Poisson distribution (Figure S10A). Per-organoid growth rates for both organoid lines decreased as the number of cells initially seeded within the microwell increased, suggesting that increased paracrine or juxtacrine signaling between cells in close proximity does not enhance growth (Figure 3B). For microwells seeded with a single-cell, single-organoid growth rates of the DKO line were higher than those of the TP53KO line (Figure 3C) and similar to growth rates observed in bulk cultures (Figure 1B). However, we observed ~ 2 -fold variability in growth rate across experiments, presumably due to differences in the composition of conditioned media across batches.⁵⁰ Despite this, DKO organoids consistently grew faster than TP53KO organoids within the same experiment (Figure 3C), establishing that the higher fold change in cell numbers for DKO organoids in bulk experiments (Figure 1B) is due to an enhanced growth rate in individual organoids rather than growth of a larger proportion of cells. Single-organoid growth rates for the same organoid line across different macrowells within the 12-well imaging plate varied only slightly, confirming lack of macrowell-specific growth effects (Figure 3D).

To calculate the time required for cells to complete a first division, we also determined the time point at which 2 cells were first identified within microwells originally seeded with a single cell. In all 3 experiments for both TP53KO and DKO mutants, some cells divided soon after initial seeding (0–24 h post-seeding), whereas the remainder took longer to divide (24–120 h post-seeding) (Figure 3E). We also observed experiment-to-experiment variation in the time to first division. For example, the TP53KO cells in all three experiments displayed a bimodal distribution in which the cells either began dividing soon after seeding or at a much later time point. However, the “slow” cells in the second replicate experiment took longer to enter cell division compared with the other two experiments. Although the DKO cells displayed similar bimodal behavior across all replicate experiments, fewer cells in the first experiment began division soon after seeding. The total distance moved exhibited a similar bimodal pattern, as cells that took longer to divide had more time for movement (Figure S10B). Total distance moved was moderately positively correlated with the time to first division ($R^2 = 0.3209$; Figure S10C) and weakly negatively correlated with growth rate ($R^2 = 0.0148$; Figure S10D). Cells tended to move the furthest immediately after seeding and then settled down with decreasing movement before dividing (Figure S10E). We further assessed if spatial confinement affects growth and found no relationship between organoid’s distance from the microwell wall and its growth rate (Figure S10F).

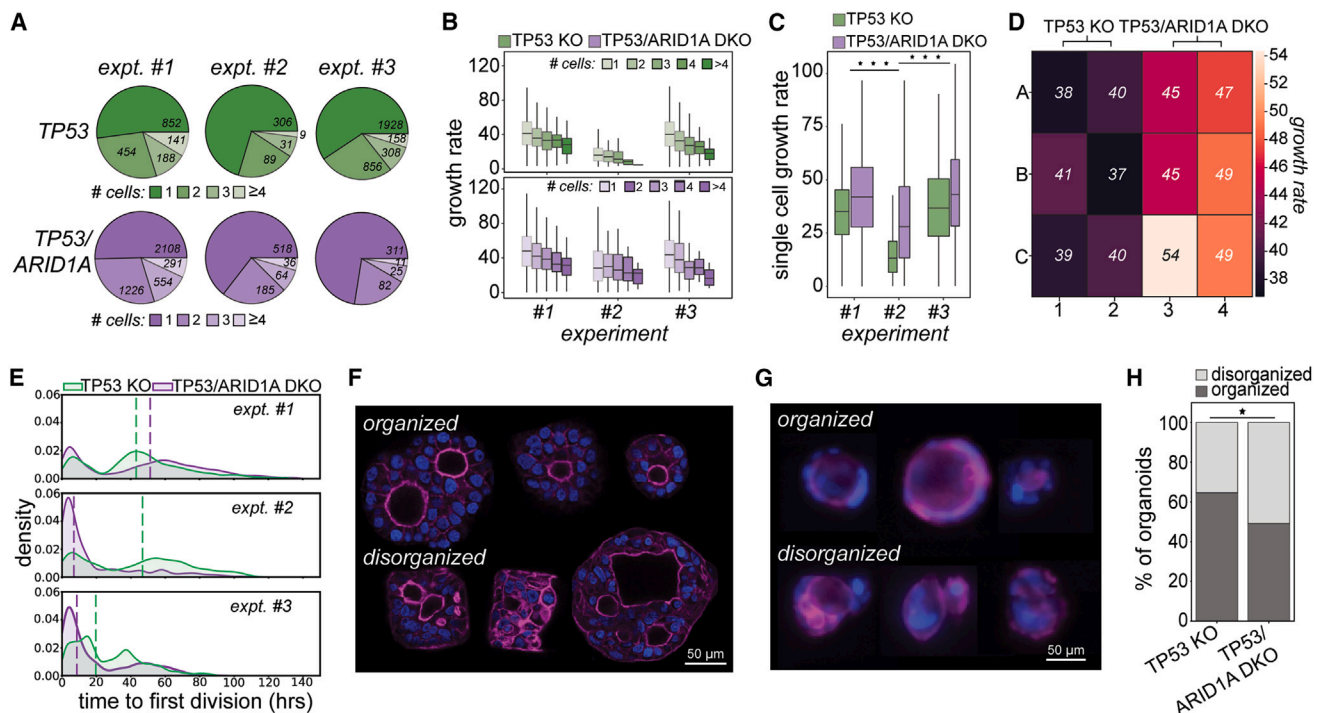


Figure 3. Microwell experiments and analysis reveal differences in growth rates and epithelial apical/basal polarization

(A) Pie charts indicating the number of TP53KO and TP53/ARID1A DKO cells loaded per microwell that exhibit growth across each experiment.

(B) Growth rates as a function of the number of cells per microwell across each experiment for TP53KO (top, green) and TP53/ARID1A DKO (bottom, purple) organoid lines.

(C) Growth rates for all single cells across each experiment for TP53KO (green; $n = 852, 306, \text{ and } 1,928$, respectively) and TP53/ARID1A DKO (purple; $n = 2,108, 518, \text{ and } 311$, respectively) organoid lines. Wilcoxon rank-sum tests yielded significant differences between experiment #2 and either experiment #1 or #3 for both within-TP53KO and within-DKO comparisons (***) denotes Bonferroni-adjusted p value < 0.001).

(D) Heatmap showing median growth rates (% daily change in cell area) for all microwells within each macrowell for experiment #1; median growth rates are indicated within each well.

(E) Kernel density plots showing the time to first cell division for microwells containing a single cell from TP53KO or TP53/ARID1A DKO organoid lines; dashed lines indicate median time to first cell division.

(F) Example confocal images of organized and disorganized polarity organoids.

(G) Example fluorescent images of organized and disorganized polarity organoids.

(H) Percentage of organoids with organized versus disorganized polarity in TP53KO and DKO lines. Fisher's exact test yielded a significant difference between the two groups (p value = 0.0007; 35.4% and 50.8% from each group, respectively).

Microwell-enabled investigation of single organoids with disordered polarity

Disordered apical-basal polarity is commonly seen in diverse epithelial cancers^{47,51} and is believed to be an early hallmark of gastric cancer.^{51–53} Intriguingly, in previous work, we noted that TP53/ARID1A DKO gastric organoids exhibit a disordered polarity.⁷ To assess whether apical-basal polarity could be visualized in organoids grown in microwells, we performed confocal imaging of both TP53KO and DKO, which confirmed that both TP53KO and DKO lines contained organoids with disordered polarity. To systematically quantify changes in membrane organization, we visually inspected the final time point fluorescence images for 257 and 236 TP53KO and TP53/ARID1A DKO cells, respectively, and classified organoids as having either classically “organized” or “disorganized” apicobasal polarity.^{54,55} Classification of disorganized polarity was based on the requirement that organoids display two of the following three characteristics: (1) disordered actin signal not restricted to organoid lumens, (2) lack of a central lumen ringed with actin, and/or (3) presence of

multiple small, disorganized lumens (Figures 3F and 3G). A greater proportion of the DKO organoids exhibited disorganized cellular phenotypes (50.8% versus 35.4% in the TP53KO organoids, $p = 0.0007$, Fisher's exact test), suggesting that either more complex engineered genotypes or loss of *ARID1A* drives greater cellular disorganization (Figure 3H).

Chromosome alterations associated with enhanced organoid growth

The high-resolution phenotyping data from this initial set of experiments (Figure 3) provided evidence that subtle differences in experimental conditions (e.g., different media batches) between experiments can drive phenotypic differences large enough to obscure meaningful intrinsic biological variation (Figure 3C). To systematically characterize variability within and across TP53KO versus DKO organoids, we performed a second set of experiments under more tightly controlled conditions (STAR Methods) in which all cells were passaged using a standardized chemically defined media (Figure 4A); these experiments used arrays of

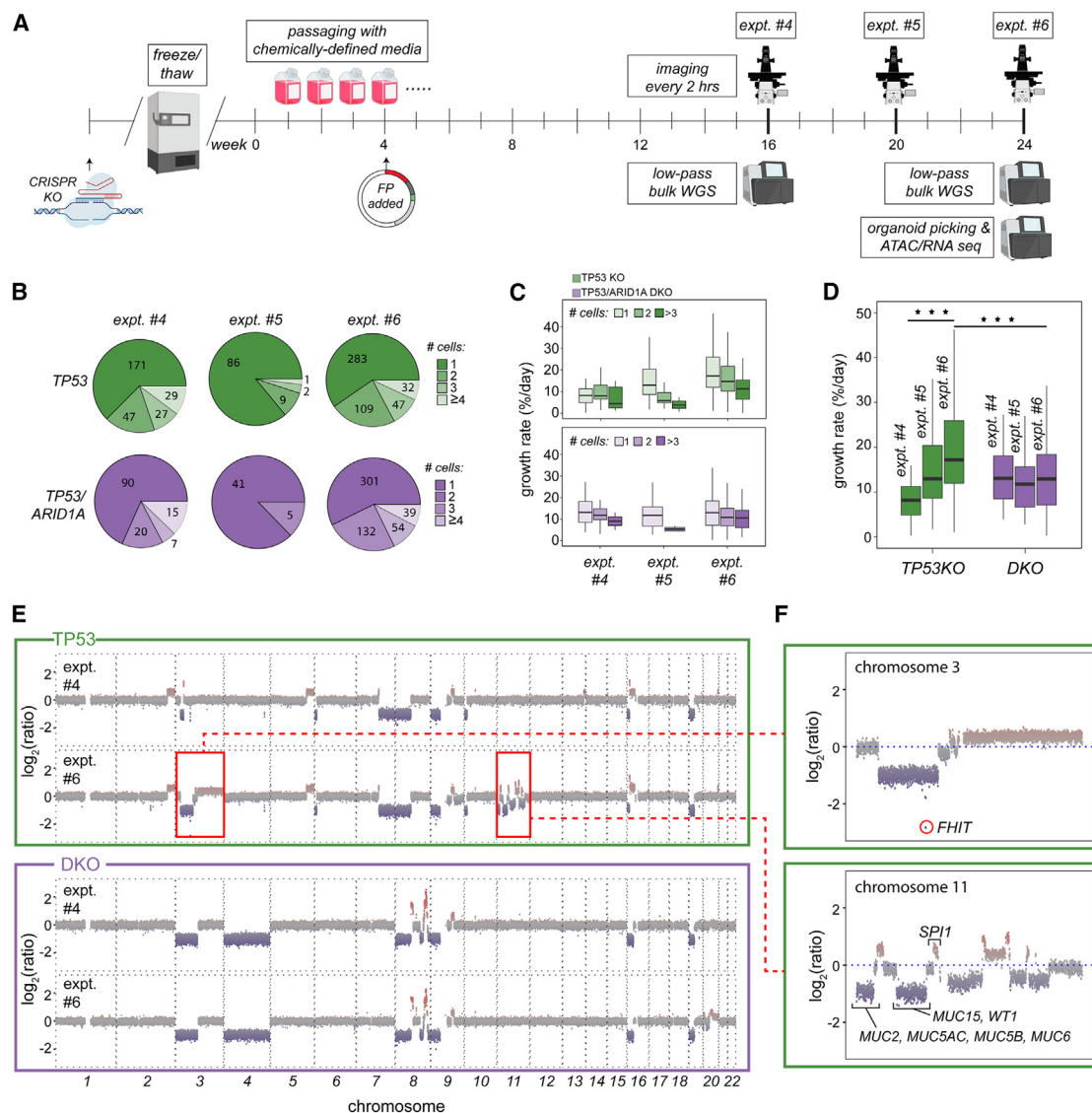


Figure 4. Microwell experiments under controlled experimental conditions reveal genomic changes associated with increased growth rates for TP53KO organoids

(A) Schematic of experiments using chemically defined growth media. TP53KO and DKO organoid lines were revived from cryopreservation and lentivirally transduced with H2B-mCherry at a defined MOI of 0.1 in 200 μ m microwells; shallow whole-genome sequencing was performed on bulk cultures at the indicated time points.

(B) Pie charts indicating the number of TP53KO and TP53/ARID1A DKO cells loaded per microwell across each experiment.

(C) Growth rates as a function of the number of cells per microwell across each experiment for TP53KO (top, green) and TP53/ARID1A DKO (bottom, purple) organoid lines.

(D) Growth rates for all single cells across each experiment for TP53KO (green; $n = 171, 86,$ and $283,$ respectively) and TP53/ARID1A DKO (purple; $n = 90, 41,$ and $301,$ respectively) organoid lines. TP53KO organoids grew increasingly faster with each experiment, whereas the growth rates of DKO organoids remained relatively constant (***) denotes Bonferroni-adjusted p value < 0.001 ; Wilcoxon ranked-sum tests).

(E) Whole-genome copy-number plot for TP53KO (green box) and DKO organoids (purple box) from experiments #4 and #6.

(F) Zoomed-in view showing copy-number variations across chromosomes 3 and 11 for TP53 organoids at the start of experiment #6.

200- μ m microwells to enable retrieval of organoids of interest using tubing with an outer diameter of 250 μ m. In total, this second set of experiments profiled 406, 144, and 997 loaded microwells that exhibited growth, respectively (Figure 4B).

As in the previous pilot experiments (Figure 3), growth rates generally decreased as per-microwell occupancy increased for both TP53KO and DKO organoids (Figure 4C). Most other im-

age-based analyses such as distance moved and time to first division also recapitulated the observations made in the pilot experiments (Figures 3 and S11). The DKO organoid growth rates remained relatively constant across experiments (Figure 4D), highlighting the importance of growth media composition for determining growth rates. However, although TP53KO organoid growth rates were initially slower than those of DKO organoids,

consistent with prior experiments, TP53KO growth rates increased steadily across experiments, eventually significantly exceeding DKO organoid growth rates (Figure 4D).

Previously, we observed that TP53KO organoids rapidly and continuously accumulate copy-number variations (CNVs) and aneuploidy.⁴³ To evaluate whether increased growth rates might be attributable to additional CNVs, we performed shallow whole-genome sequencing (sWGS) of both TP53KO and DKO organoid cells frozen prior to the start and at the end of the second set of experiments. The sWGS for the TP53KO organoids revealed additional large-scale copy-number alterations on chromosome 3 at the end of the experimental timeline (labeled experiment #6 in the figure for clarity), as well as apparent chromosome shattering encompassing all of chromosome 11, that were not present previously (labeled experiment #4 in the figure) (Figure 4E). By contrast, DKO cells showed no significant CNV changes across experiments (Figure 4E), corroborating the previous observation that DKO organoids did not exhibit molecular signatures associated with the chromosomal unstable subtype of gastric cancer but instead displayed transcriptional profiles characteristic of the microsatellite unstable subtype.⁷

Next, we investigated regions affected by CNVs in TP53KO cells to identify specific genetic changes that could explain the observed increase in growth rate. Although the TP53KO organoids harbored a focal deletion of FHIT locus early in the second set of experiments (seen in passage #9 and #13 post-thaw), a common early alteration in this model and gastric cancers,⁴³ the entire chromosome 3p arm was deleted at the later time point (Figure 4F). FHIT is a secondary regulator of DNA damage response commonly altered in TP53-null organoids⁴³ and gastric cancer,^{56,57} and these data suggest that the loss of both FHIT and TP53 likely drives rapid accumulation of additional CNVs. On chromosome 11, we observed loss of several regions containing multiple mucin genes (MUC2, MUC5AC, MUC5B, MUC6, and MUC15) (Figure 4F) that are often dysregulated during malignant progression of gastrointestinal cancers.^{58–60} For instance, abrogation of MUC5AC expression has been associated with increased cell proliferation and vascular invasion in gastric tumor cells.^{61,62} In addition, we observed amplification of *SP11*, which is a proto-oncogene upregulated in various cancers and implicated in proliferation^{63–65} (Figure 4F). These alterations potentially explain the increased growth rate observed in the TP53KO organoids over time.

Sequencing of disorganized versus organized polarity organoids reveals changes in chromatin accessibility associated with morphology and adhesion

To explore the molecular changes driving differences in cell polarity, we extracted DKO organoids of interest from the microwell arrays for downstream sequencing using a syringe pump and 3D-printed microscope adapter (STAR Methods; Figure 5A). After staining organoids in microwells at the end of experiment #6 with a live-cell actin dye (STAR Methods), we retrieved ten organoids with organized polarity and ten with disorganized polarity (based on the criteria described in the STAR Methods) from the DKO line (Figures 5B and S12). We then prepared libraries for single-organoid dual ATAC-seq and RNA sequencing (RNA-seq) to identify differentially accessible regions (DARs) or differentially expressed genes (DEGs), respectively, that might explain differences in polarity.

Given the limiting biomass derived from single organoids, we used a dual library preparation protocol developed for low-input specimens.⁶⁶ As the data quality was generally higher for ATAC-seq libraries compared with RNA-seq libraries, we initially focused on the ATAC-seq data. Following read deduplication and exclusion of 6 organoid samples with <60,000 uniquely mapped reads or <500 accessible ATAC-seq peaks, we profiled 7 organized and 7 disorganized polarity DKO gastric organoids with an average of 191,000 unique ATAC-seq reads and 4,425 accessible peaks per sample (Figure 5C). Clustering based on ATAC-seq DARs clearly separated organoids with organized versus disorganized polarity (Figure 5D), and the DARs were enriched for gene ontology (GO) terms related to cell adhesion and morphogenesis (Figure 5E). Principal-component analysis (PCA) also revealed a clear separation between organized and disorganized polarity organoids (Figure 5F). At a molecular level, transcription factor (TF) footprinting analysis of the ATAC-seq data indicated that organoids with disorganized polarity had increased accessibility in chromosomal regions bound by the SP-family TFs (Figure 5G), which are involved in maintaining epithelial cell polarity^{67,68} and are dysregulated in a variety of cancer types.^{69,70} Consistent with this potential role, the DNA binding footprints of SP-family TFs were enriched in upstream promoter regions of several genes such as *PPP2R1B* and *LLGL1* involved in establishing epithelial apical-basal polarity (Figure 5H).^{68,71,72} The RNA-seq data corroborated these findings, as multiple upregulated genes with known SP-family target sites were differentially expressed in organized versus disorganized polarity organoids (Figures 5I and S13). These data thus suggest a plausible molecular basis for disorganized polarity in TP53/ARID1A DKO gastric organoids and demonstrate the feasibility and utility of molecularly characterizing single organoids with distinct imaging-based phenotypes.

DISCUSSION

Here, we describe an easy-to-use, open-source microwell platform for high-throughput, image-based phenotyping of organoids with the ability to retrieve single organoids of interest for additional downstream molecular profiling (e.g., single-organoid ATAC-seq or RNA-seq). As a first demonstration of this platform, we used an organoid-optimized deep-learning model to identify and track nearly 100,000 individual cell trajectories from 2 engineering human gastric organoid model lineages within >8,000,000 microwell images. Using these data, we then quantified organoid growth rates and positions over time with high accuracy and granularity. These measurements led to the identification of specific molecular features associated with increased organoid growth and loss of apical-basal polarity.

This microwell platform offers multiple advantages over other existing organoid culture methods. In comparison with bulk culture, the microwell method combined with our custom open-source 3D image analysis allows precise measurement of quantitative phenotypes for individual organoids, (e.g., single-organoid growth rates, division times, and migration distances) rather than bulk averages (e.g., fold change in cell number for an entire population).⁷³ Unlike commercially available methods or closed microfluidic systems,^{15,33} our microwell arrays do not require specialized instrumentation and are thus easy to adapt to existing

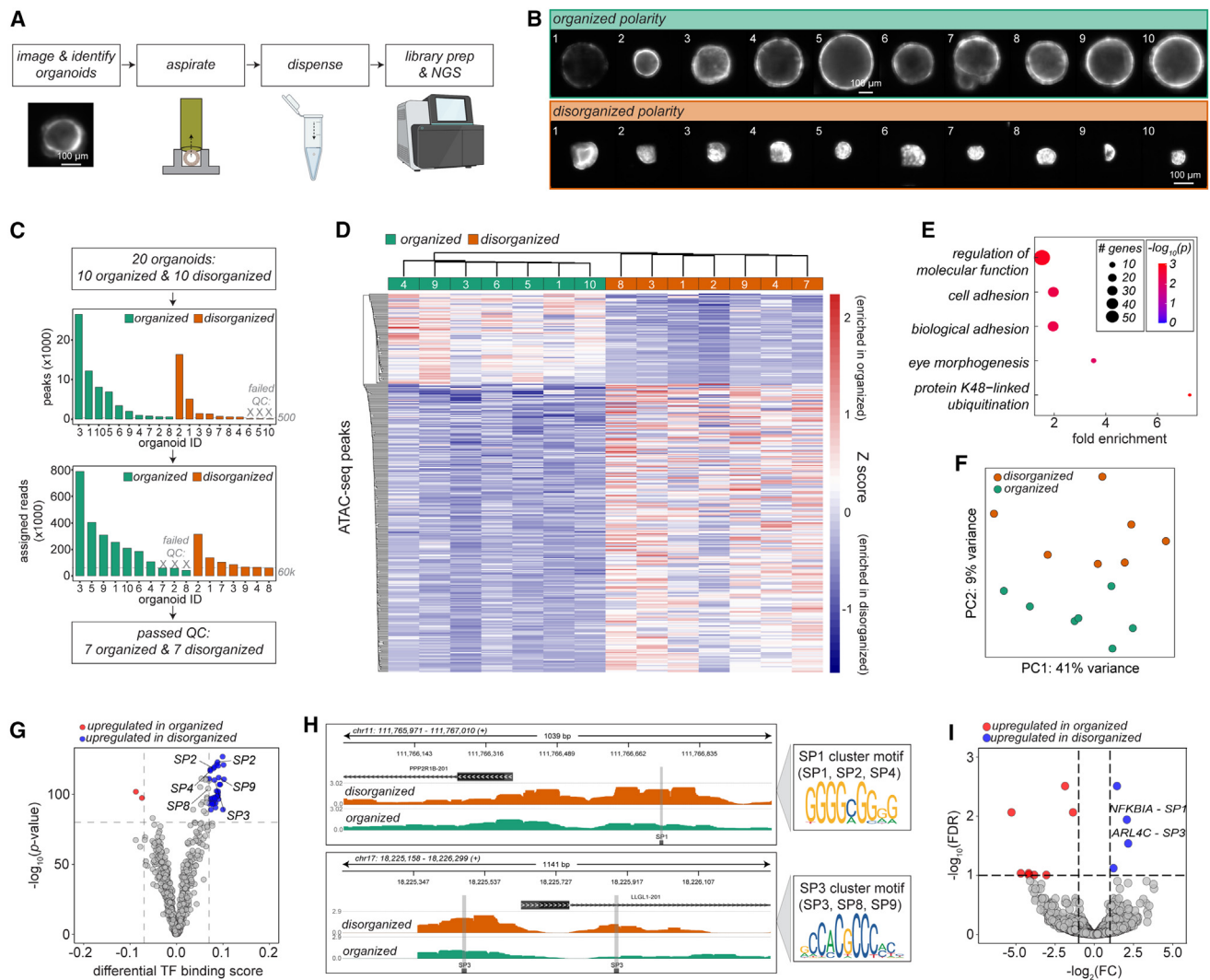


Figure 5. Single-organoid retrieval and sequencing reveals changes in chromatin accessibility associated with changes in apical/basal polarity

(A) Schematic of experimental pipeline for single-organoid retrieval. Single organoids were retrieved using a syringe pump connected to a 3D-printed microscope adapter via polyether ether ketone (PEEK) tubing and dispensed into a clean vial prior to preparing ATAC-seq and mRNA-seq libraries.

(B) Images of all retrieved organoids. Top: organoids with organized polarity (i.e., a robust actin ring around the lumen). Bottom: organoids with disorganized polarity (i.e., disorganized actin signals). Scale bar in bottom right applies to all organoids except top #5, which has its own scale bar.

(C) ATAC-seq quality control pipeline eliminating samples with insufficient read depths or peaks.

(D) Heatmap showing differentially accessible peaks between organized and disorganized organoids.

(E) Gene ontology analysis of differentially accessible peaks showing enrichment of the peaks in pathways associated with cellular adhesion and morphology. The color scale represents $-\log_{10}(p)$ values.

(F) Principal-component analysis of ATAC-seq data showing clear separation between organized and disorganized organoids.

(G) Transcription factor footprinting analysis of ATAC-seq data showing enriched binding of SP-family transcription factors in disorganized organoids.

(H) Footprinting plots showing increased binding of SP-family transcription factors in PPP2R1B and LLGL2 regulatory regions involved in regulating apical-basal polarity in epithelial cells.

(I) Volcano plot showing differentially expressed genes in RNA-seq data. Two of the four upregulated genes in disorganized organoids are known to harbor binding sites for SP-family transcription factors.

cell culture workflows.²⁹ In addition, we generated complete organoids from single cells seeded in microwells,^{34–37} a particularly relevant advance for studying the early stages of tumorigenesis in which a single transformed cell develops into a tumor. Our microwell platform has comparatively higher throughput than other existing methods, profiling thousands of organoids in parallel as

opposed to hundreds at a time.^{27,36,41} Finally, our microwell platform also facilitates retrieval of individual organoids of interest, which can then be either directly sequenced to investigate the molecular changes driving particular phenotypes or clonally expanded *in vitro* to generate sufficient biomass for additional phenotypic or molecular profiling.

In future work, we anticipate that the ability to profile single organoids will enable personalized drug screening in patient-derived tumor organoids. Specifically, this platform offers the potential to screen patient-derived tumor organoids against a wide variety of drugs using much lower input materials. Moreover, the organoid retrieval mechanism described here makes it possible to retrieve resistant organoids and either characterize them via “omics” approaches or test them against additional drug candidates.

Future modifications to our approach have the potential to unlock multiple additional phenotyping capabilities. For example, combining more frequent imaging intervals (e.g., on the order of minutes rather than hours) with the 3D-trained neural network could make it possible to track cell divisions and cell lineages⁴⁵ for fast-dividing cell types, extending our ability to measure migration to multiple cells and enhancing resolution of cell division time measurements. In addition, loading of cells tagged with a different fluorescent protein could allow deeper investigation into the molecular mechanisms driving the observation that microwells loaded with a greater number of cells tended to grow more slowly. For example, the addition of fluorescent tags on proteins such as MUC5AC and PGC (commonly used as cell type markers in mucous cells and chief cells, respectively) could reveal whether certain cell types are likely to promote or suppress growth. Similarly, we could test whether adjacent apoptotic cells slow the growth of nearby cells by fluorescently labeling Annexin V.⁷⁴ Finally, we could visualize chromosome missegregation errors during cell division using our H2B-mCherry nuclear reporter with higher magnification imaging.

In summary, we present a microwell-based 3D culture platform that can be easily adapted for the quantitative characterization of various image-based phenotypes. We use this platform to quantify cell-to-cell variability in growth rate and apical-basal polarity in two engineered gastric organoid models and discover potential molecular mechanisms underlying the variability in both phenotypes. Through our findings, we demonstrate the importance of single-organoid measurements for characterizing morphological and molecular changes associated with disease progression, which are not feasible in bulk culture and using traditional measurement approaches.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cels.2023.08.002>.

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AUTHOR CONTRIBUTIONS

A.S., W.W., K.K., C.C., and P.M.F. conceptualized the initial research idea. A.S. and W.W. performed the microwell experiments. A.S. tested and made the microwell devices and performed image-processing bioinformatic analysis. W.W. performed organoid-picking experiments, made single-organoid sequencing libraries, and performed bioinformatic analysis. S.L. and D.M. set up the microscope, performed initial testing, and implemented the organoid-picking platform. T.V. performed manual labeling of deep-learning training and test data. V.C. performed initial growth and cell seeding experiments. D.V.V., J.S., and A.S. set up the deep-learning pipeline, and P.M.F. wrote code to quantify cell growth from processed images. K.K. and Y.-H.L. performed the organoid gene editing. A.S., W.W., C.C., and P.M.F. wrote the manuscript.

DECLARATION OF INTERESTS

Unrelated to this study, C.C. is a stockholder in Illumina and DeepCell, and an advisor to DeepCell, Genentech, Bristol Myers Squibb, 3T Biosciences, NanoString, and ResistanceBio. Unrelated to this study, C.K. is a founder and stockholder for Surrozen Inc, Mozart Therapeutics, and NextVivo. Unrelated to this study, P.M.F. is an advisor to Evoxyme. C.C. and P.M.F. are on the advisory board of the journal. W.W., A.S., C.C., and P.M.F. have filed a provisional patent (Patent Application No. 63/380.196) related to this work.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Villin-1 mouse anti-human, Alexa Fluor 647	Novus Biologicals	NBP2-50081AF647; RRID:AB_2943649
CK8 rabbit anti-human, Alexa Fluor 647	Abcam	ab53280; RRID:AB_869901
Bacteria and virus strains		
pLenti6-H2B-mCherry	Addgene	#89766
Chemicals, peptides, and recombinant proteins		
Advanced DMEM/F12 basal medium	Gibco	12634010
Penicillin-Streptomycin (100X)	Gibco	15140122
Penicillin-Streptomycin-Glutamine (100X)	Gibco	10378016
HEPES buffer (1M)	Gibco	15630080
GlutaMax (100X)	Gibco	35050061
TrypLE Express solution	Gibco	12563029
Normocin	Invivogen	ant-nr-1
1H,1H,2H,2H-perfluorooctyl-trichlorosilane	Sigma-Aldrich	448931
N-acetyl L-cysteine	Sigma-Aldrich	A9165-100G
Gastrin	Sigma-Aldrich	G9145-5MG
N21-MAX	R&D System	AR008
FGF10	R&D System	345-FG
Human recombinant Noggin	R&D System	6057-NG
Human recombinant R-spondin1	R&D System	4645-RS
Y-27632	Tocris	1254/10mg
CHIR 99021	Tocris	4423/10mg
SB202190	Tocris	1264
A83-01	Tocris	2939
EGF	Peptrotech	AF-100-15
Wnt-FC surrogate protein	Immunoprecise	N001-0.5mg
Cultrex BME Type 2	Bio-Techne	3532-005-02
RTV 615	R.S. Hughes	RTV615 CLEAR 010
BD Cytofix/Cytoperm Fixation/ Permeabilization solution	BD Biosciences	554722Tr
BD Perm/Wash buffer	BD Biosciences	554723
SiR-actin kit	Cytoskeleton	CY-SC001
Critical commercial assays		
Qiagen Allprep DNA/RNA Mini Kit	Qiagen	80204
Illumina Nextera XT DNA Library Prep Kit	Illumina	FC-131-1024
Dynabeads mRNA Direct Micro Kit	Invitrogen	61021
Qiagen MinElute PCR Purification Kit	Qiagen	28004
Superscript IV First Strand cDNA Synthesis Kit	Invitrogen	18091050
Q5 High-Fidelity 2X Mastermix	NEB	M0492S
AMPure XP beads	Beckman Coulter	A64881
10X Chromium Single Cell 3' Reagent Kit	10X Genomics	PN-1000268
Deposited data		
Microwell images and processed files (OSF)	This paper	OSF: https://doi.org/10.17605/OSF.IO/R4S3M
NGS data and processed files (dbGAP)	This paper	dbGAP: phs003315.v1.p1

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Cell lines		
TP53KO human gastric organoid line	Lo et al. ⁷	TP53KO
TP53/ARID1A DKO human gastric organoid line	Karlsson et al. ⁴³	DKO
Software and algorithms		
AutoCAD	Autodesk	https://www.autodesk.com/products/autocad/
DeepCell	Van Valen et al. ⁴⁴ Moen et al. ⁴⁵	https://github.com/vanvalenlab/deepcell-tf https://github.com/vanvalenlab/deepcell-toolbox
BWA	Li and Durbin ⁷⁵	https://github.com/lh3/bwa
Samtools	Li et al. ⁷⁶	http://samtools.sourceforge.net
QDNaseq	Scheinin et al. ⁷⁷	https://github.com/ccagc/QDNaseq
Nextflow	Di Tommaso et al. ⁷⁸	https://www.nextflow.io
MACS2	Zhang et al. ⁷⁹	https://github.com/mac3-project/MACS
HOMER	Heinz et al. ⁸⁰	http://homer.ucsd.edu/homer/
DESeq2	Love et al. ⁸¹	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
TOBIAS	Bentsen et al. ⁸²	https://github.com/loosolab/TOBIAS
Trim-Galore	Krueger et al. ⁸³	https://github.com/FelixKrueger/TrimGalore
STAR	Dobin et al. ⁸⁴	https://github.com/alexdobin/STAR
RSEM	Li and Dewey ⁸⁵	https://github.com/deweylab/RSEM
Revigo	Supek et al. ⁸⁶	http://revigo.irb.hr
Seurat	Butler et al. ⁸⁷	https://satijalab.org/seurat/
Codes for MicrowellProcessor (github)	This paper	https://doi.org/10.5281/zenodo.8193787
Codes for MicrowellStitcher (github)	This paper	https://doi.org/10.5281/zenodo.8193799
Codes for wellception (DeepCell) (github)	This paper	https://doi.org/10.5281/zenodo.8210528
Codes for plots associated with NGS data (github)	This paper	https://doi.org/10.5281/zenodo.8201767
Other		
SU-8 2100 photoresist	Microchem	SU-8 2100

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Polly Fordyce (pfordyce@stanford.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

All sequencing raw data, including shallow whole-genome sequencing, dual bulk ATAC/RNA sequencing, and single-cell RNA sequencing data have been deposited at dbGAP and are publicly available as of the date of publication. The accession number is provided in the [key resources table](#). Other data (image files for all experiments; labeled training, validation, and test set data for deep learning model training; processed data files; and per-experiment and per-microwell summary reports) have been deposited at OSF and are publicly available as of the date of publication. The DOI is provided in the [key resources table](#).

All original code has been deposited on GitHub and is publicly available as of the date of the publication. The DOIs are listed in the [key resources table](#).

Any additional information required to reanalyze the data reported in this paper is available from the [lead contacts](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

CRISPR-engineered gastric organoids

The P53KO and TP53/ARID1A DKO gastric organoids were generated in and obtained from previous studies.^{7,43} Briefly, non-malignant gastric tissue from the corpus (stomach body) was obtained during sleeve gastrectomy of a male patient at Stanford University Hospital under an IRB-approved protocol. Wild-type gastric organoids were established followed by CRISPR/Cas9 mediated knockout of *TP53* (P53KO).⁴³ CRISPR/Cas9-mediated *ARID1A* knockout (KO) in primary *TP53*^{-/-} human gastric organoids yielded double knockout lines (TP53/ARID1A DKO).^{7,43} These two organoid lines, representing pre-malignant and malignant states, were used in two separate sets of experiments. In the first set (experiments #1 - #3), we conducted proof-of-principle tests to determine various types of measurements that we could perform by culturing single-cell derived gastric organoids in the microwells. Prior to microwell experiments, organoids were cultured in 24-well tissue culture plates using growth media containing 50% Wnt3A/R-spondin1/Noggin conditioned media produced in house, 50% Advanced DMEM/F12, 1X Penicillin/Streptomycin/Glutamine, 1X Normocin, 1X B-27 Supplement, 1X GlutaMax, 1mM N-Acetyl-L-cysteine, 500nM A83-01, 10uM SB202190, 10mM Gastrin, and 50ng/mL EGF. The media was further supplemented with 10μM Y-27632 and 2.5μM CHIR-99021 during passaging to promote stem cell survival. Culture media was refreshed every 7 days and organoid cultures were passaged once every 12-14 days. During each passage, old media was removed and 500ul of TrypLE was added to each well to dissolve the Cultrex BME and dissociate organoids into single cells. After 30-40 minutes of incubation at 37°C, cells were pelleted down by centrifuging at 500g for 5 minutes. The supernatant was then removed, and the cell pellet was resuspended in wash media (Advanced DMEM/F12, 1X HEPES, 1X Penicillin/Streptomycin/Glutamine) for cell counting. Cell counting was performed by combining 10ul of cell suspension with 10μl Trypan Blue and then loading the mixture into a Countess chip for automated counting using the Countess II Cell Counter (Thermo Fisher). An appropriate volume of cells was transferred to a new tube, spun down at 500g for 5 minutes, and the pelleted cells were resuspended in fresh Cultrex BME. Cells were re-plated in a new 24-well plate with 20000 cells per 40μl of Cultrex BME dome. The plate was then incubated at 37°C for 20 minutes to allow Cultrex BME to solidify; after incubation, 500μl fresh growth media was added. The second set of experiments (#4 - #6) used chemically defined complete growth media to prevent media batch effects from confounding downstream analysis. The chemically defined media was composed of Advanced DMEM/F12, 100ng/mL Noggin, 1000ng/mL R-spondin1, 0.5nM Wnt-FC surrogate protein, 1X Penicillin/Streptomycin, 1X Normocin, 1X N21-Max Supplement, 1X GlutaMax, 1mM N-Acetyl-L-cysteine, 500nM A83-01, 10uM SB202190, 10mM Gastrin, and 50ng/mL EGF. During passaging, 10μM Y-27632, 2.5μM CHIR-99021 and 200ng/ml FGF10 were added. All other culture conditions remained the same as described in the previous section.

METHOD DETAILS

Lentiviral transduction to fluorescently tag organoids with histone H2B protein

Addgene plasmid #89766 expressing H2B-mCherry fusion protein was packaged into lentiviral particles by the Gene Vector Virus Core at Stanford University. Prior to transduction, organoids were washed with PBS and incubated with TrypLE at 37°C for 40 minutes. FBS was then added to quench the TrypLE reaction. After that, the dissociated cells were centrifuged at 500g for 5 minutes and resuspended in growth media supplemented with 10μM Y-27632. The H2B-mCherry lentiviral particles were added at an MOI of 0.1 to aliquots of 500k cells to maximize the number of successfully transduced cells harboring just one copy of the transgene, thereby reducing insertional mutagenesis and theoretically maintaining a constant fluorescence intensity. The cell/virus suspension was transferred to a single well of a 24-well plate, and a 1-hour spinoculation at 600g at 32°C was performed. After spinoculation, cells were incubated at 37°C for four hours before being dissociated and pelleted down at 500g for 5 minutes. After pelleting, the pellet was resuspended in Cultrex BME followed by plating onto a 24-well plate. Organoids were allowed to recover for 3-5 days until mCherry expression was visible, then transduced cells were FACS-sorted for cells expressing positive mCherry signal. Sorted cells were allowed to recover and expand for several passages before being used for experiments.

Microwell design and fabrication

Design files for molds containing arrays of square microwells with a variety of different dimensions (from 100 x 100 x 80μm to 1000 x 1000 x 80μm; breath x width x height) were generated in AutoCAD and printed on standard transparencies at 30000 dpi (Fineline Imaging). All design files are available in [supplemental information](#). Molds were created from SU-8 2100 photoresist (Microchem, Inc.) on a 4" silicon test-grade wafer (University Wafer) according to the manufacturer's instructions. After fabrication, molds were treated with vapor deposition of 1H,1H,2H,2H-perfluorooctyl-trichlorosilane (Sigma Aldrich) under vacuum for 10 minutes. Microwell devices were made from the molds using standard soft lithography (Figure S3). Briefly, RTV 615 precursor solutions at a ratio of 10:1 (base elastomer:curing agent) (R.S. Hughes Co.) were mixed using a THINKY mixer with 3 minutes of mixing followed by 3 minutes of degassing, both at 2000 rpm, poured onto molds, and spun on a spin coater (Laurell Technologies) at 200 rpm with an acceleration of 133 rpm/s for 30 seconds to spread the PDMS and create a layer of PDMS approximately 0.5 mm thick (Figure S4); this thickness could be easily peeled off but remained thin enough for high-quality imaging using a long working distance objective. The PDMS was then degassed in a vacuum chamber for 15 minutes and baked at 80°C for 20 minutes. After baking, the PDMS was peeled from the mold and cut into arrays of square devices (Figure S3). Prior to use, the devices were treated with 20% oxygen plasma for 8 minutes,

placed at the bottom of 12-well plates with forceps, sterilized overnight by immersion in 70% ethanol, then treated with 0.5% PBS-BSA for >1 hour to render the devices hydrophilic for cell growth.

Cell loading into microwells

Dissociated organoids obtained during bulk culture passage were resuspended in WENR media to a concentration of 6000 cells/ml for 100 μ m microwells and between 600–2000 cells/ml for 200 μ m microwells. The cell suspension was pipetted directly onto BSA-treated microwell arrays, and plates were centrifuged at 500g for 5 min to load cells into microwells. Following cell loading, excess media was aspirated and Cultrex BME was pipetted dropwise onto the microwell arrays containing cells. Plates were incubated at 37°C to polymerize the Cultrex BME, then growth media supplemented with 10 μ M Y-27632 and 2.5 μ M CHIR-99021 and 200 ng/ml FGF10 was added.

Time-lapse image acquisition

For initial bright field imaging experiments (Figure 1), organoids in microwells were grown in a standard tissue culture incubator and imaged 1X per day over 7–10 days using a Keyence BX-700 microscope. A grid acquisition with a 10X objective and 20X final magnification was performed using the built-in brightfield capability of the Keyence microscope.

High-resolution time-lapse images (Figures 2, 3, and 4) were acquired on an inverted fluorescence microscope (Nikon Ti) with a motorized xy-stage (ASI MS-2000) and an sCMOS camera (Andor Zyla 4.2+) set to acquire images at 2x2 binning for a final resolution of 1024x1024. Broad spectrum and fluorescence illumination was provided by a solid-state light source (Lumencor Sola). Images were acquired with a 10X objective at 20x final magnification. For multi-day acquisitions, organoids were kept in a stage-top incubator set to maintain 37°C, 5% CO₂, and 95% relative humidity. Imaging was controlled with a Jupyter notebook that used a custom Python library (AcqPak) to manage experimental acquisitions and the Micro-Manager API for hardware control; all AcqPak software is freely available for download from Github (<https://github.com/FordyceLab/AcqPak>). Each position in a rastered acquisition was imaged with the following filter cubes and exposures: brightfield (Semrock BRFD-A-NTE-ZERO) at 1 ms, Cy5 (Semrock 49002) at 15 ms, and mCherry at 50ms. Rasters were acquired every 2 hours.

Initial image processing

All image processing was performed using custom Python libraries which are openly available for download from Github. To extract per-microwell information, we first stitched raw tiled images into a single reconstructed image of the microwell array (MicrowellStitcher; <https://github.com/FordyceLab/MicrowellStitcher>). We then used recorded pixel locations of the array corners to rotate the image so the array edges are parallel with the image edges. Subarrays containing either 400 microwells (100 μ m microwells) or 100 microwells (200 μ m microwells) were extracted from the stitched array image using the corner pixel locations (MicrowellProcessor; <https://github.com/FordyceLab/MicrowellProcessor>), and image prediction for nuclear segmentation was performed with a trained deep learning model on the subarrays (Wellception; <https://github.com/juliaschaepe/wellception>).

Deep learning model training and testing

To enable automated quantification of organoid growth from microwell images, we first trained a deep learning model^{44,45} to predict fluorescent organoid nuclei. To construct the dataset used to train the model, individual microwell images of 114x114 pixels were saved as .npz files and each image was randomly assigned to either training data (80%) or validation data (20%). Additional images from the first set of experiments were used as test data (306, 348, 258 images respectively). We used automated quality control to remove microwell images containing >8 cells after initial testing revealed images with higher cell numbers were challenging for annotators to label manually (data not shown), leaving a final dataset of 2711 training images, 542 validation images, and 912 test images. The Caliban desktop module⁴⁵ was used to manually label individual nuclei in each image and the annotations were saved to the .npz files. We then used deepcell-tf (<https://github.com/vanvalenlab/deepcell-tf>) to train a deep learning model, with model weights stored in a hdf5 file, and performance was measured on the validation and test sets using the deepcell-toolbox (<https://github.com/vanvalenlab/deepcell-toolbox>).^{44,45}

To predict nuclei in microwell images, the organoid segmenter was initialized with trained model weights stored in an hdf5 file. The segmenter took in input images and preprocessed them with histogram normalization using a kernel size of 32 x 32. Each microwell was imaged in 9–15 subarrays, with each 2280 x 2280 pixel subarray containing 400 microwells of 114 x 114 pixels each (100 μ m microwells) or 100 microwells of 224 x 224 pixels each (200 μ m microwells). Each subsection was first rescaled by a factor of two before being passed into the model. Model outputs were postprocessed with a watershed filter with a detection threshold of 0.25, a distance threshold of 0.1, and a minimum distance of 2.5. The model was run on a NVIDIA GPU. Nuclear predictions were saved as an hdf5 file containing all subarrays coming from a single microwell array. A.csv file containing summary statistics of the predictions for each timepoint and microwell (indexed by pixel location) was generated that included the predicted number of nuclei, area of each nucleus, the total area of the organoid, centroid locations for each organoid, and pertinent metadata such as timepoint, microwell ID, mutant, and microwell pixel location.

Quantification of organoid growth rates using deep learning

To quantify organoid growth rates, we first used the extracted deep learning predictions to identify microwells initially loaded with ≥ 1 cells for which the model identified $\geq n+1$ cells at the final timepoint in the time course. Total predicted organoid nuclear area was

plotted as a function of time for up to 60 timepoints to include only timepoints with >8 cells. A model of exponential growth was then fit to the data, and parameters corresponding to initial area and growth rate as the percent change in area per day were extracted from the exponential model. To compare growth rate distributions across mutants and experiment, we performed a one-way ANOVA with three variables and applied a Bonferroni correction to account for multiple hypothesis testing.

Polarity measurements with confocal imaging

Initial confocal imaging to assess apical-basal polarity was performed in microwells. Growth media was aspirated from plates, arrays were washed with 1X phosphate-buffered saline (PBS), the arrays were incubated in BD Cytofix/Cytoperm Fixation/Permeabilization solution for 30min at 4°C, and then cells were washed 3x with 1X BD Perm/Wash buffer, with 10 minute incubations at room temperature between washes. Fixed cells were then stained with DAPI at 300 nM concentration diluted in PBS solution and Alexa Fluor 647 phalloidin at 165 nM concentration diluted in PBS solution, then incubated at room temperature in the dark for 1 hr. Cells were washed with PBS solution before imaging. Imaging was performed with a Leica SP5 upright multi-photon confocal microscope with a HCX APO L20x/1.00 water immersion objective. Organoid images were acquired in a z-stack with images taken every 10 μ m. Representative single-plane images were extracted using Velocity software.

Polarity measurements with live-cell fluorescence imaging

Old growth media was aspirated from microwells at the final timepoint of the time course imaging experiments, and growth media supplemented with 1X SiR-actin dye and 1X verapamil was added to macrowells. This was done to reduce the extrinsic effects introduced by the addition of the actin dye on the growth measurements of the organoids during time course imaging. Imaging, image processing, and individual microwell extraction were performed as described above. Blinded manual polarity classification was performed on a subset of images from the final timepoint of experiment #1, with 408 microwells classified for TP53KO and 403 microwells for DKO. Organoids were classified as “organized”, “disorganized”, or “unknown”. To be classified as “disorganized”, organoids were required to display 2 of the following 3 characteristics: (1) disorganized actin signal not restricted to organoid lumens, (2) lack of a central lumen ringed with actin, and/or (3) the presence of multiple small, disorganized lumens. Aspiration of organoids with varying polarity was performed at the end of the second set of experiments, using the same classification requirements.

Shallow WGS of bulk organoids

The P53KO and DKO organoids were harvested using TrypLE solution, similar to organoid passaging, prior to the start of experiments #4 and #6. The cells were then lysed and the nucleic acid was extracted using Qiagen Allprep DNA/RNA Mini Kit (Qiagen). Aliquots of DNA were sent to Novogene Co. for the construction of sequencing libraries and shallow WGS at 1X coverage. Sequencing reads were aligned to the hg38 human reference genome using BWA.⁷⁵ Samtools⁷⁶ was used to convert the alignment files into bam format with indexes. The bam files were subsequently analyzed with QDNAseq⁷⁷ to infer copy-number variations using 50kb read bin size and median normalization.

Single organoid retrieval from microwell

A syringe pump (Harvard Apparatus Pump 11 Elite) was used to drive a 1 ml syringe fitted with a blunt Luer probe tip. The syringe was attached to 255 μ m ID/510 μ m OD PEEK tubing via a short 1 cm Tygon splint. The PEEK tubing was then threaded through a tightly-fitting 20 μ l pipette tip such that approximately 0.5cm extended past the tip. The splint and tip connections were secured with super glue. The tip was affixed to the microscope condenser z-stage using a 3D-printed liquid light guide holder (Figure S12). The syringe and tubing were primed with phosphate-buffered saline (PBS). The touch-down position of the tip was adjusted to the center of a live field of view under 4X magnification using thumbscrews on the condenser slot; this position was noted by drawing a reference box around the tip location in the Micro-Manager GUI. To pick an organoid of interest, the xy-stage was moved until the organoid was centered in the reference box. The tip was then lowered to the microwell surface using the condenser z-stage. The syringe pump was then used to withdraw the organoid into the tip; if necessary, adherent organoids were loosened by toggling withdraw/infuse on the syringe pump and/or by incubation with TrypLE for 5 minutes. The tip containing the aspirated organoid was then positioned just above an empty container (e.g. a well of a multiwell plate) and dispensed using the infuse button on the syringe pump into a 1.5ml centrifuge tube. A total of 20 organoids were retrieved from the DKO culture in microwells. Ten organoids had organized apical basal polarity as evidenced by the actin ring with a large lumen (Figure 4B). Ten organoids had disorganized apical basal polarity as defined throughout based on the presence of two of the three criteria: (1) disorganized actin signal not restricted to organoid lumens, (2) lack of a central lumen ringed with actin, and/or (3) the presence of multiple small, disorganized lumens.

Dual ATAC/mRNA library preparations and sequencing of single retrieved organoids

Each single organoid retrieved from microwell was subjected to dual ATAC/mRNA sequencing library preparation following a recently published protocol that was designed for low input.⁶⁶ Briefly, the organoid was spun down at 500g for 5 minutes using a benchtop centrifuge. After removing supernatant, the organoid was then resuspended in a direct permeabilization/tagmentation mastermix (25 μ l TD buffer (from Illumina Nextera XT DNA Library Prep Kit), 2.5 μ l Tn5, 16.5 μ l DPBS, 0.5 μ l 1% digitonin, 0.5 μ l 10% Tween-20, 2.5 μ l RNase inhibitor and 2.5 μ l nuclease-free water) without prior nuclei isolation. The reaction mixture was incubated in a 37°C water bath for 30 minutes with occasional hand-pipetting. At the end of incubation, 2.5 μ l of stop buffer containing 10mM EDTA and 0.5M lithium chloride was added to neutralize the permeabilization/tagmentation reaction. The tagmented cells were then lysed using

100 μ l of Lysis/Binding Buffer from the Dynabeads mRNA Direct Micro Kit (Invitrogen). After lysis, 20 μ l of pre-washed Dynabeads Oligo-dT beads were added to the lysate which was thereupon incubated at room temperature for 5 minutes to allow mRNA to anneal to the Oligo-dT. The beads with annealed mRNAs were separated from the supernatant using a magnetic rack. The supernatant which contained genomic DNA was transferred to a new tube for subsequent genomic DNA extraction using Qiagen MinElute PCR Purification Kit. Meanwhile, the beads-mRNA complex was resuspended in 20 μ l of reverse transcription mastermix (without any primer) from Superscript IV First Strand cDNA Synthesis Kit (Invitrogen). The Oligo-dT on beads served as primers for the reverse transcription reaction. The reaction mix was incubated at 50°C for 5 minutes then at 55°C for an additional 10 minutes. The resulting cDNA/mRNA hybrid was thus covalently bound to the magnetic beads. The beads were washed twice with 100 μ l ice-cold 10mM Tris-HCl (pH 7) and resuspended in 5 μ l of Tris-HCl. Sequencing library preparation of the cDNA was performed on beads using Nextera XT DNA Library Prep Kit where the reagent volumes were halved (keeping the reagent ratio consistent). Meanwhile, the previously tagged ATAC-DNA was amplified using Q5 High-Fidelity 2X Mastermix (NEB) with universal i5 and indexed i7 adapters. PCR was performed with 18 cycles due to the low input nature of the samples. Both ATAC and cDNA libraries were cleaned using AMPure XP beads according to manufacturer's recommendations. The libraries were then sent to Novogene Co. for sequencing on the Illumina Novaseq platform with 75bp paired-end reads.

Bioinformatic analysis of ATAC-seq libraries

The raw sequencing reads were first processed with Trim-Galore to remove Illumina adapter sequences.⁸³ The Nextflow atacseq pipeline (nf-core/atacseq) was used to process the trimmed reads with default parameters.⁷⁸ Briefly, reads were aligned to hg38 human reference genome using BWA.⁷⁵ The alignment files were further processed with Picard and Samtools to mark duplicate reads and to create bigWig files for downstream analysis and data visualization.⁷⁶ Open chromatin peaks were called using MACS2⁷⁹ and annotated with HOMER.⁸⁰ Due to low sample input, the number of PCR cycles used to generate the library was more than the recommended number, and this resulted in higher read duplication rate. As a result, the data was subjected to additional filters: we removed 1) samples with less than 60000 deduplicated, uniquely mapped reads, and 2) samples with less than 500 total peaks. The final dataset consisted of seven organoids each with organized and disorganized apical-basal polarity (Figure 5C). The peak regions were then analyzed in DESeq2 to determine differential accessibility regions (DARs).⁸¹ The transcription factor footprinting analysis was performed using TOBIAS with default parameters.⁸² The DARs were subjected to gene ontology (GO) analysis using default parameters.^{88–90} The GO terms with a p-value <0.01 were retained for Revigo analysis to remove redundant GO terms.⁸⁶

Bioinformatic analysis of mRNA-seq libraries

The raw sequencing reads were first processed with Trim-Galore⁸³ to remove Illumina adapter sequences. The trimmed reads were then aligned to hg38 human reference transcriptome using STAR,⁸⁴ and processed using RSEM to calculate gene expressions in each sample.⁸⁵ The data was further examined with PCA analysis to remove outlier samples. In addition, the genes with zero read-count in more than or equal to 30% of the samples were removed from the differential analysis to ensure that the analysis was not skewed by missing data. DESeq2 was used to determine differential gene expression between the organized and disorganized groups.⁸¹ Gene ontology analysis was performed in similar manner as described for ATAC-seq libraries with one modification: GO terms with a p-value <0.05 were retained for Revigo analysis due to a low number of differentially expressed genes (Figure S13). Single-cell RNA sequencing was also performed on organoids that were seeded either onto microwell or in bulk using 10X Chromium Single Cell 3' Reagent Kit. The 10X sequence outputs were processed and visualized with Seurat.⁸⁷

Statistical Analysis

Details for each statistical test and analysis can be found in the results section and the figure legends (i.e. number of organoids and types of tests). Wilcoxon rank-sum test is used in pairwise comparisons of organoid growth rates between different genetic backgrounds. Fisher's exact test is used in pairwise comparisons of phenotypic traits.